Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)	A passing photon with energy equal to 1.79 eV (or equivalent) (1) Causes another photon {of same frequency / same energy / in phase} to be emitted (1) accept similar photon emitted	2			2		
		(ii)	So stimulated emission is more {frequent / probable / likely} than absorption	1			1		
	(b)		$\lambda = \frac{hc}{\Delta E}$ **or** $f = \frac{\Delta E}{h}$ **and** $\lambda = \frac{c}{f}$ transposition at any stage (1) $\Delta E = 2.86 \times 10^{-19}$ J or by implication (1) $\lambda = 694$ n[m] (1) **ecf** on ΔE		3		3	2	
	(c)	(i)	Number per second= $\frac{0.60}{2.86\times10^{-19}\textbf{ecf}}$ **or** 2.1×10^{18} [s^{-1}]		1		1	1	
		(ii)	$p = \frac{6.63\times10^{-34}}{6.94\times10^{-7}\textbf{ecf}}$ or by implication **ecf** (1) $p = 9.6 \times 10^{-28}$ N s or kg m s^{-1} **unit mark** (1)		2		2	2	
		(iii)	Momentum per second $=$ Answer to (i) $\times$ Answer to (ii) **or** 2.0×10^{-9} [N] (1) Force on reflecting surface = $2 \times 2.0 \times 10^{-9} = 4.0 \times 10^{-9}$ [N] **ecf** (1) Award 1 mark only for multiplication by 2 anywhere		2		2	2	
			Question 9 total	**3**	**8**	**0**	**11**	**7**	**0**